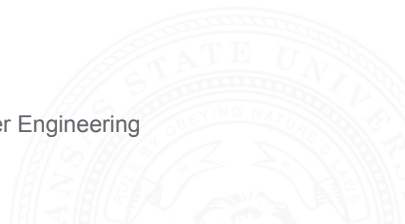


Exam 1 Review

Phasors, three-phase circuits, transformer tests, and
per unit

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Review outline

- 1 Part 1: RMS phasors and complex power.
- 2 Part 2: balanced three-phase circuits and feeder calculations.
- 3 Part 3: transformer tests and per-unit calculations.
- 4 Part 4: solution checks. Optional review: planning arithmetic.

The core follows the electrical topics through L14 reviewed in L16. Planning pages are optional extension.

Starting problem

STARTING PROBLEM

A calculation has the right formula but the wrong RMS value, winding side, or base. Which early declaration prevents each error?

Declare the voltage reference, RMS convention, physical side, and per-unit bases before substituting numbers.

PART 1

Phasors and complex power

RMS phasors

$$x(t) = \sqrt{2}X_{\text{rms}} \cos(\omega t + \theta) \quad \Longleftrightarrow \quad \mathbf{X} = X_{\text{rms}} \angle \theta$$

Time domain

$\sqrt{2}X_{\text{rms}}$ **is peak**

The coefficient is not the RMS magnitude.

Phasor domain

X_{rms} **is RMS**

The common frequency is suppressed, not lost.

Declare the cosine/RMS convention once, then preserve it throughout the solution.

Time domain to RMS phasors

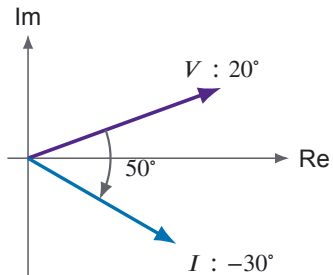
$$v(t) = 240\sqrt{2}\cos(377t + \pi/9) \text{ V}$$

$$i(t) = 10\sqrt{2}\cos(377t - \pi/6) \text{ A}$$

$$V = 240\angle 20^\circ \text{ V}$$

$$I = 10\angle(-30^\circ) \text{ A}$$

$$\theta_V - \theta_I = 50^\circ$$



V and A use separate scales

RMS conversion changes magnitude and preserves phase displacement. Current lags voltage here.

Complex-number forms

Add or subtract

$$120\angle 0^\circ + 80\angle 90^\circ = 120 + j80$$
$$= \boxed{144.22\angle 33.69^\circ}$$

Multiply or divide

$$\frac{A\angle\alpha}{B\angle\beta} = \frac{A}{B}\angle(\alpha - \beta)$$

Convert back only when the next operation requires it.

A calculator cannot repair an operation performed in the wrong coordinate form.

Complex power uses the current conjugate

$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = P + jQ$$

$$P > 0$$

real P absorbed

W, kW, or MW

$$Q < 0$$

capacitive Q supplied

leading contribution

$$Q > 0$$

inductive Q absorbed

var, kvar, or Mvar

pf

append leading or lagging

ratio $P/|S|$



Impedance power check

Review example: $\mathbf{V} = 240\angle 0^\circ \text{ V}$, $Z = 8 + j6 \ \Omega$

Current

$$Z = 10\angle 36.87^\circ \ \Omega$$
$$\mathbf{I} = \frac{\mathbf{V}}{Z} = \boxed{24\angle(-36.87^\circ) \text{ A}}$$

Complex power

$$\mathbf{S} = \mathbf{VI}^* = \boxed{4.608 + j3.456 \text{ kVA}}$$
$$\mathbf{S} = |I|^2 Z$$

$Q > 0$ and $\text{pf} = 0.800$ **lagging** agree with an inductive impedance.

Power-factor correction

0.800 to 0.950 lagging

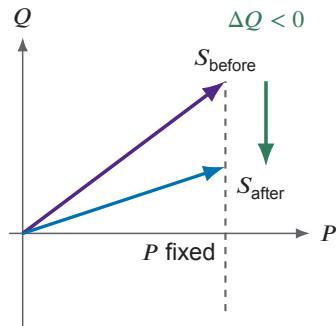
$$P = 4.608 \text{ kW}, \quad Q_1 = 3.456 \text{ kvar}$$

$$Q_2 = P \tan(\cos^{-1} 0.95) = 1.515 \text{ kvar}$$

$$Q_C = Q_2 - Q_1 = -1.941 \text{ kvar}$$

$$C = \frac{|Q_C|}{\omega V^2} = 89.4 \text{ } \mu\text{F}$$

$$|I_2| = \frac{P/0.95}{240} = 20.21 \text{ A}$$



Reactive power decreases at fixed real power. The corrected load remains lagging.

PART 2

Balanced three-phase circuits

Positive-sequence set

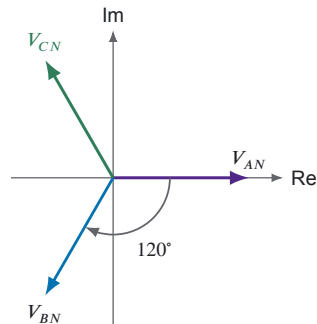
$$V_{AN} = V_p \angle 0^\circ$$

$$V_{BN} = V_p \angle (-120^\circ)$$

$$V_{CN} = V_p \angle 120^\circ$$

$$V_{AN} + V_{BN} + V_{CN} = 0$$

Equal magnitudes and 120° spacing form a balanced set.



Swapping any two phase labels reverses the sequence.

Wye line voltage

Positive sequence, $\mathbf{V}_{AN}$ reference

$$\mathbf{V}_{AB} = \mathbf{V}_{AN} - \mathbf{V}_{BN} = \sqrt{3} \mathbf{V}_{AN} \angle 30^\circ$$

$$\mathbf{I}_A = \mathbf{I}_{\text{phase},A}$$

13.8-kV bus

$$V_{\text{phase}} = \frac{13.8}{\sqrt{3}} = \boxed{7.967 \text{ kV}}$$

Line voltage is the equipment rating;
phase voltage drives the wye branch.



Delta line current

Balanced positive sequence

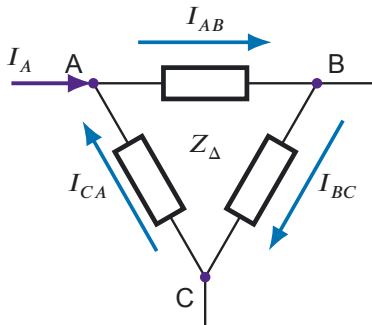
$$V_{\text{phase}} = V_{LL}$$

KCL at terminal A gives

$$I_A + I_{CA} = I_{AB}$$

$$I_A = I_{AB} - I_{CA}$$

$$I_A = \sqrt{3}I_{AB}\angle(-30^\circ)$$



The branch arrows determine the signs in the terminal-current equation.

Three-phase power

Valid for balanced wye or balanced delta

Complex-power route

$$\mathbf{S}_{3\phi} = 3\mathbf{V}_{\text{phase,rms}}\mathbf{I}_{\text{phase,rms}}^*$$

This preserves the power angle and the sign of Q .

Magnitude route

$$|S_{3\phi}| = \sqrt{3}V_{LL,\text{rms}}I_{L,\text{rms}}$$

$$P = |S| \cos \phi, \quad Q = |S| \sin \phi$$

A project MW or Mvar entry is a three-phase total, not one phase.

Example: feeder data

Receiving load

$$V_{LL,L} = 13.8 \text{ kV}$$

$$Z_{\Delta} = 60 + j45 \text{ } \Omega/\text{leg}$$

Feeder and reference

$$Z_{\ell} = 0.15 + j0.45 \text{ } \Omega/\phi$$

- Balanced positive abc sequence.
- Use $V_{AN,L}$ as the zero-angle reference.

Find line current, required sending voltage, load power, and feeder loss.

Example: solution route

1. Given

Receiving line voltage, delta load, and per-phase feeder impedance.

2. Model

Convert the balanced delta load to its wye equivalent.

3. Calculation

Find phase current, add the series drop, and compute total complex powers.

Check the line/phase factors and $S_S - S_L = 3|I_L|^2 Z_\ell$.

4. Check

The per-phase circuit contains one feeder impedance and one equivalent load impedance.

Example: voltage reference

$$\mathbf{V}_{AN,L} = \frac{13.8 \text{ kV}}{\sqrt{3}} \angle 0^\circ = \boxed{7.96743 \angle 0^\circ \text{ kV}}$$

$$\mathbf{V}_{AB,L} = \sqrt{3} \mathbf{V}_{AN,L} \angle 30^\circ = \boxed{13.8 \angle 30^\circ \text{ kV}}$$

The line-to-line magnitude is given; the wye-equivalent phase circuit requires $\mathbf{V}_{AN,L}$.

Example: delta equivalent

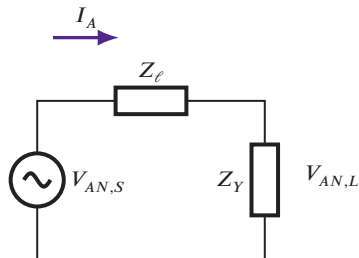
$$Z_Y = \frac{60 + j45}{3}$$

$$Z_Y = 20 + j15 \, \Omega$$

$$Z_Y = 25 \angle 36.87^\circ \, \Omega$$

$$Z_\ell = 0.15 + j0.45 \, \Omega$$

$$V_{AN,L} = 7.96743 \angle 0^\circ \, \text{kV}$$



One phase of the balanced network

$$Z_Y = Z_\Delta/3$$

The equivalent Y preserves terminal currents and total power. The line impedance remains in series.

Example: line current

$$I_A = \frac{V_{AN,L}}{Z_Y} = \boxed{318.697 \angle (-36.87^\circ) \text{ A}}$$

Rectangular form

$$I_A = (254.958 - j191.218) \text{ A}$$

Supports the feeder-voltage addition.

Original delta branch

$$I_{AB} = \boxed{184.000 \angle (-6.87^\circ) \text{ A}}$$

$$I_A = \sqrt{3} I_{AB} \angle (-30^\circ).$$

The current angle matches the impedance angle with the opposite sign.

Example: sending voltage

$$\Delta \mathbf{V}_A = \mathbf{I}_A \mathbf{Z}_\ell = \boxed{124.292 + j86.048 \text{ V}}$$

$$\mathbf{V}_{AN,S} = \mathbf{V}_{AN,L} + \Delta \mathbf{V}_A = 8092.183 \angle 0.609^\circ \text{ V}$$

$$V_{LL,S} = \sqrt{3} |\mathbf{V}_{AN,S}| = \boxed{14.0161 \text{ kV}}.$$



Example: total load power

$$\mathbf{S}_L = 3\mathbf{V}_{AN,L}\mathbf{I}_A^* = \boxed{6.09408 + j4.57056 \text{ MVA}}$$

Magnitude

$$|S_L| = \boxed{7.61760 \text{ MVA}}$$

$\sqrt{3}V_{LL,L}I_L$ gives the same value.

Power factor

$$\text{pf} = \frac{6.09408}{7.61760} = \boxed{0.800}$$

Positive Q means lagging.

The receiving real power, 6.09408 MW, is the planning peak used in the installed-capacity check.

Example: feeder loss

$$\mathbf{S}_\ell = 3|I_A|^2 Z_\ell = \boxed{0.045706 + j0.137117 \text{ MVA}}$$

$$\mathbf{S}_S = 3\mathbf{V}_{AN,S}\mathbf{I}_A^* = \boxed{6.13979 + j4.707677 \text{ MVA}}$$

$\mathbf{S}_S - \mathbf{S}_L - \mathbf{S}_\ell = 0$ to displayed precision.

PART 3

Transformer tests and per unit

Transformer quantities

Ideal winding relations

$$a = \frac{N_H}{N_L} = \frac{|V_H|}{|V_L|}, \quad \frac{|I_H|}{|I_L|} = \frac{1}{a}$$

$$Z_{L \rightarrow H} = a^2 Z_L$$

Physical model

- The leakage branch represents winding resistance and leakage reactance.
- The shunt branch represents core loss and magnetizing current.
- Record the side to which each parameter is referred.

Changing the referred side changes the numerical impedance while preserving the same physical transformer.

Practical transformer model

Series branch: copper loss and leakage.

$$Z_{eq} = R_{eq} + jX_{eq}$$

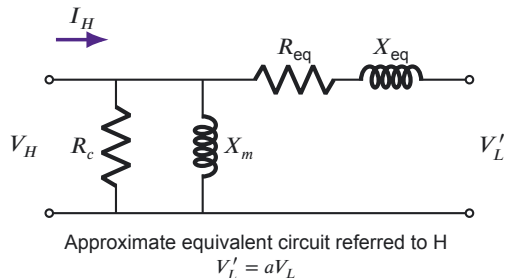
Shunt branch: core loss and magnetization.

$$R_c \parallel jX_m$$

OC test: rated voltage, small current.

SC test: reduced voltage, rated current.

Regulation compares no-load and full-load voltage at fixed source.



Approximate equivalent circuit referred to H

$$V'_L = aV_L$$

The approximate referred circuit separates the two loss mechanisms.

Example: short-circuit test

Single-phase equipment

$$S_{\text{rated}} = 50 \text{ kVA}$$

$$V_H/V_L = 2000/200 \text{ V}$$

Test on H side

$$V_{SC} = 100 \text{ V}, \quad I_{SC} = 25.0 \text{ A}$$

$$P_{SC} = 500 \text{ W}$$

Find the H-referred series resistance and leakage reactance. The test current is rated H current.

Example: test solution

$$|Z_{eq,H}| = \frac{V_{SC}}{I_{SC}} = 4.00 \, \Omega$$

$$R_{eq,H} = \frac{P_{SC}}{I_{SC}^2} = 0.800 \, \Omega$$

$$X_{eq,H} = \sqrt{4.00^2 - 0.800^2} = 3.919 \, \Omega$$

The measured wattage determines resistance. The measured voltage/current ratio determines impedance magnitude.

Single-phase per-unit bases

50-kVA equipment base

$$Z_{B,H} = \frac{2000^2}{50000} = 80.0 \, \Omega$$

$$Z_{B,L} = \frac{200^2}{50000} = 0.800 \, \Omega$$

One physical leakage branch

$$z_{50} = \frac{0.800 + j3.919}{80.0}$$

$$\approx 0.0100 + j0.04899$$

**For this single-phase record, use single-phase VA and winding terminal volts.
Compatible H/L bases preserve the same per-unit impedance.**

Practice: referral and base

Practice

Use the same 2000/200-V transformer and its H-referred $Z = 0.800 + j3.919 \, \Omega$.
Change to a 100-kVA base with unchanged voltage bases.

- 1 Find actual L-referred impedance.
- 2 Find new per-unit impedance from either physical side.
- 3 Explain why the 50-kVA rating remains the equipment rating.

Practice: referral answer

$$Z_{\text{eq,L}} = \frac{0.800 + j3.919}{10^2} = 0.00800 + j0.03919 \, \Omega$$

$$z_{100} = z_{50} \frac{100}{50} \approx 0.0200 + j0.09798$$

$$Z_{B,H,100} = 40.0 \, \Omega, \quad Z_{B,L,100} = 0.400 \, \Omega$$

Base conversion changes the numerical representation. Equipment MVA remains the basis for its loading limit.

PART 4

Solution checks

Review error checks

- 1 Phasors: use RMS magnitude, the declared cosine reference, and the current conjugate.
- 2 Connections: carry the positive-sequence angle and line/phase factors.
- 3 Transformer tests: keep the measured side and the referred side explicit.
- 4 Per unit: put old and new voltage bases on the same physical side.
- 5 Conservation: close total load, modeled loss, and delivered source power.

A final value needs a visible calculation chain and an independent check.

Core review summary

Topic	Calculation route	Independent check
AC power	$S = VI^*$	$S = I ^2 Z$
Three phase	one phase, then total	$ S = \sqrt{3} V_{LL} I_L$
Transformer test	$V/I, P/I^2$	rated test current
Per unit	actual value / base	reconstruct actual ohms

The electrical core covers the same phasor, three-phase, transformer, and per-unit conventions used in L16.

PART 5

Planning extension

Optional review after the electrical core

Demand, energy, capacity

Quantity	Typical unit	Meaning
Demand	MW	Instantaneous rate of electrical use
Energy	MWh	Demand accumulated over time
Installed rating	MW	Installed maximum rating
Capacity factor	%	Actual energy divided by rated-energy potential
Planning margin	% of peak	Extra installed capacity above forecast peak

MW is a rate; MWh is a time integral. They cannot be added or substituted.

Load-duration curve

Chronological demand: 4, 6, 3, 5 MW.

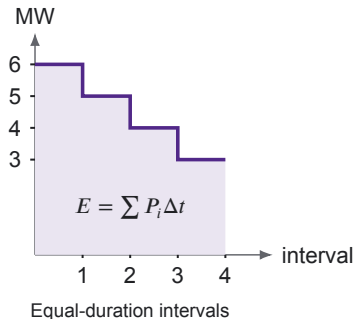
Sorted demand: 6, 5, 4, 3 MW.

$$P_{\text{peak}} = 6.0 \text{ MW}$$

$$P_{\text{inst,min}} = 1.15(6) = 6.90 \text{ MW}$$

For equal interval duration Δt ,

$$E = (6 + 5 + 4 + 3)\Delta t.$$



Sorting preserves energy when each demand retains its represented duration.

Screening curves

Annual cost per installed kW

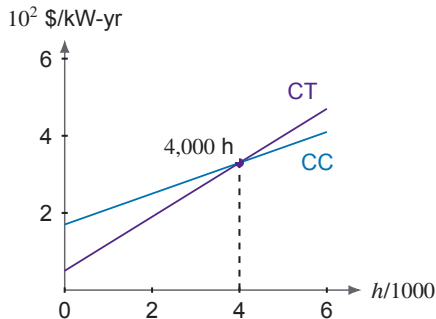
$$C(h) = F + \frac{vh}{1000}$$

$$C_{CT} = 50 + 0.070h$$

$$C_{CC} = 170 + 0.040h$$

$$h^* = 4000 \text{ h/year}$$

CT is cheaper below the crossover. CC is cheaper above it.



The intercept represents fixed cost. The slope represents variable cost per operating hour.

Renewable shares

$$f_r = \frac{P_{\text{wind}} + P_{\text{solar}}}{P_{\text{inst}}},$$

$$f_w = \frac{P_{\text{wind}}}{P_{\text{inst}}}, \quad f_s = \frac{P_{\text{solar}}}{P_{\text{inst}}}.$$

Total renewable

$$f_r \geq 30\%$$

Combined installed share.

Wind

$$f_w \geq 10\%$$

Separate installed share.

Solar

$$f_s \geq 10\%$$

Separate installed share.

The renewable subtotal is the numerator, never the denominator.

Feasible generation mix

Installed-capacity tests

- integer counts for every technology;
- total nameplate at least 115% of peak;
- renewable, wind, and solar shares all pass.

Peak-available test

$$P_{\text{avail}} = \sum_k a_k n_k P_{k,\text{unit}}$$

$$P_{\text{avail}} \geq P_{\text{req}}$$

Nameplate adequacy and peak-available adequacy are different inequalities.

Planning extension: data

- Use the preceding feeder load as the planning peak: 6.09408 MW.
- Portfolio: five 1.10-MW gas units, two 1.00-MW wind units, and one 1.00-MW solar unit.
- Peak availability: gas 100 percent, wind 40 percent, solar 10 percent.
- Apply the 15-percent nameplate margin and existing 30/10/10-percent renewable-share requirements.

Installed-nameplate shares and availability-weighted power use different records.

Planning: installed capacity

$$P_{\text{installed}} = 5(1.10) + 2(1.00) + 1(1.00)$$

$$P_{\text{installed}} = \boxed{8.50 \text{ MW}}$$

$$P_{\text{min}} = 1.15(6.09408) = 7.008 \text{ MW}$$

Renewable

$$\geq 30\%$$

$$3/8.5 = 35.29\%$$

Wind

$$\geq 10\%$$

$$2/8.5 = 23.53\%$$

Solar

$$\geq 10\%$$

$$1/8.5 = 11.76\%$$

The calculated receiving load supplies the planning peak. Compare installed capacity and all three renewable shares with their requirements.

Planning: peak adequacy

$$P_{\text{avail}} = 5.50 + 0.40(2.00) + 0.10(1.00) = \boxed{6.400 \text{ MW}}$$

$$6.400 - 6.13979 = \boxed{0.260214 \text{ MW available}}$$

Planning

8.50 MW nameplate

passes 115% of receiving peak

Operation

6.400 MW available

covers source real power including feeder loss



Planning: result summary

Planning

- 8.50 MW installed;
- 35.29% renewable;
- 6.400 MW peak-available;
- 0.260 MW operating headroom.

Balanced feeder

- $I_L = 318.697 \text{ A}$;
- $V_{LL,S} = 14.0161 \text{ kV}$;
- $P_L = 6.09408 \text{ MW}$;
- $P_\ell = 0.045706 \text{ MW}$.

A defensible solution states its reference, units, connection, total-versus-phase meaning, and final conservation check.